Congruence and Modular Arithmetic

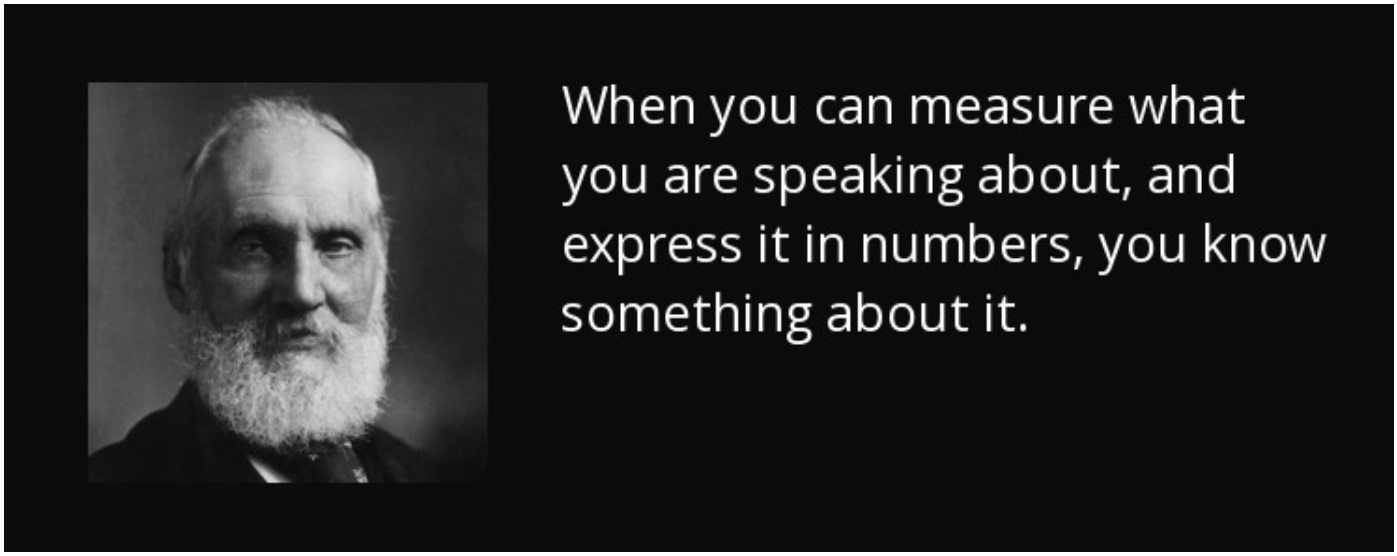
Worksheet 6

Student Name: _____ **Student ID:** _____

This lecture will help you understand

- quotient, remainder, and the meaning of modulo,
- standard remainders and negative representatives,
- cyclic patterns in modular arithmetic,
- remainder families and congruence modulo n,
- properties of congruence and reduction of large calculations.

> *"I see, I forget.*
> *I hear, I remember.*
> *I do, I understand."*
>
> — Confucius



Quick Revision: GCD

1. Two numbers have
$$\gcd(a, b) = 12, \qquad \text{lcm}(a, b) = 420.$$
If one number is 60, find the other.

Answer: _____

2. Use the Euclidean Algorithm to find $\gcd(252, 198)$.
Complete:
$$252 = 198(___) + ____$$
$$198 = ____(___) + ____$$
$$____ = ____(___) + ____$$
$$\therefore \gcd(252, 198) = _____$$

The Machine in the Basement

In this worksheet, we will explore division and remainders through a conversation between Ant and Gnam. As they interact with the strange machine, follow their questions and complete the tasks along the way.

Warm-up Problem!



One evening, Ant and Gnam find an old locked basement under the Mathematics Lab. Inside the basement, they see a strange machine. On the machine, something is written:

"Feed me a number. I will tell you what is left behind."

Gnam: "What does that mean?"
Ant: "It sounds like a remainder machine."
Gnam: "Remainder? Like division?"
Ant: "Exactly. But this machine will take us much deeper than ordinary division."

The machine suddenly opens a small tray. Inside it are chocolates and empty boxes.
Machine: "Each box holds exactly 5 chocolates. Put 17 chocolates inside me."
Gnam puts 17 chocolates into the machine. The machine asks them to complete its first message:

$$17 = 5 \times ____ + ____$$

Full boxes = _____ Leftover chocolates = _____

The machine makes a soft sound and prints more number-trials. Ant says, "Let us record the full boxes and the leftovers each time."

Activity 1: Quotient and Remainder

Number	Divisor	Quotient	Remainder	Division Form
17	5			$17 = 5(___) + ___$
23	7			$23 = 7(___) + ___$
36	8			$36 = 8(___) + ___$
108	3			$108 = 3(___) + ___$
129	7			$129 = 7(___) + ___$

After looking at the machine's answers, Ant asks:
What do you notice about the remainder in each case?

Your Answer:

After the warm-up, let us make our understanding of division more precise by recalling!

Divide each number by 5 and write the remainder.

$$5 \div 5 \quad \Rightarrow \quad \text{Remainder} = ____$$

$$6 \div 5 \quad \Rightarrow \quad \text{Remainder} = ____$$

$$7 \div 5 \quad \Rightarrow \quad \text{Remainder} = ____$$

$$8 \div 5 \quad \Rightarrow \quad \text{Remainder} = ____$$

$$9 \div 5 \quad \Rightarrow \quad \text{Remainder} = ____$$

So, when dividing by 5, the possible remainders are :

____, ____, ____, ____, ____.

Can we get any other possible remainder when dividing by 5?

☐ Yes ☐ No

Therefore, when dividing by 5, the remainders go from

$$0, 1, ..._____.$$

Now think generally.
When a number is divided by n, the possible remainders are

$$0,\ 1,\ 2,\ \ldots,\ _____.$$

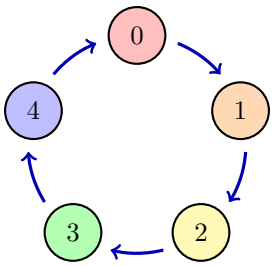
These become the n positions of our cycle.

Scene 1: The Circular Track

The machine lights up. A circular track appears on the floor with five positions:

$$0, 1, 2, 3, 4.$$

After 4, the track returns to 0.



Gnam asks: "Why are there only five positions? Where did the other numbers go?"

Ant explains: "They did not disappear. They are walking around the same cycle again and again. This is the idea of working **modulo** 5: divide by 5, ignore the complete rounds, and keep only the final position or remainder."

Definition

Modulo n means remainders after division by n. Equivalently, it means working on a cycle with n positions:

$$0, 1, 2, \ldots, n-1.$$

So modulo 5 means we only care about one of the five positions $0, 1, 2, 3, 4$.

Starting from 0, Gnam walks 17 steps around the track.

$$17 = 5(3) + 2.$$

After every 5 steps, he comes back to 0. After 15 steps, he is back at 0, and 2 more steps take him to position 2.

So, when Gnam walks 17 steps, he lands on position 2.

Now try a few more walks and check where Gnam lands each time.

Use the cycle $0 \to 1 \to 2 \to 3 \to 4 \to 0$.
Complete the table.

Steps	Division Form	Landing Position
7	$7 = 5(___) + ___$	
12	$12 = 5(___) + ___$	
18	$18 = 5(___) + ___$	
23	$23 = 5(___) + ___$	
31	$31 = 5(___) + ___$	

What do you notice?

The landing position is the same as the _____.

Scene 2: The Strange Negative Walk

Gnam now walks backwards on the same modulo 5 cycle.

Starting at 0, complete the landing positions.

1) Walking 1 step backward from 0 lands on _____.
2) Walking 2 steps backward from 0 lands on _____.
3) Walking 3 steps backward from 0 lands on _____.
4) Walking 3 steps backward reaches the same position as walking _____ steps forward.

Gnam asks: "Can walking backward also describe a position on the cycle?"

Ant says: "Let us check this with a number."

Now look at 17.
First write 17 using the usual division form:

$$17 = 5(___) + ___$$

So the landing position of 17 on the modulo 5 cycle is _____.

Now look at the next multiple of 5 after 17.

$$20 = 5(___)$$

How many steps backward from 20 gives 17?

_____

So we can also write

$$17 = 5(___) - ___$$

Now compare the two completed forms:

$$17 = 5(3) + 2 \qquad \text{and} \qquad 17 = 5(4) - 3.$$

1) Do $+2$ and -3 give the same landing position on the modulo 5 cycle?

☐ Yes ☐ No

2) If we describe the landing position by walking forward from 0, which number do we use?

☐ 2 ☐ -3

3) In division language, this forward landing position is called the standard remainder.
So the standard remainder is _____.

4) But -3 is also useful. Instead of saying "2 steps forward," it describes the same landing position as "3 steps _____."

Gnam says: "Let me check if I understood this game on another cycle."

Ant says: "Good. Try the modulo 7 cycle now."

1) On the modulo 7 cycle, 10 lands at the same position as _____.
2) On the modulo 7 cycle, -3 lands at the same position as _____.
3) On the modulo 7 cycle, -21 lands at the same position as _____.
4) On the modulo 7 cycle, -23 lands at the same position as _____.
5) On the modulo 7 cycle, 25 lands at the same position as _____.
6) On the modulo 7 cycle, -15 lands at the same position as _____.

Ant says: "Good. You now know that different numbers can land at the same position on a cycle. Let us see how a whole city uses this idea."

Scene 3: The Gate of Titan



The machine opens a portal. Ant and Gnam enter Titan, a city where arithmetic wraps around.

At the gate, a guard says:

"Only citizens with remainder badge 3 *modulo* 11 *may enter."*

Gnam shows his card: 25. The guard allows him inside.

Gnam asks: "But my card says 25, not 3!"

The guard replies: "On Titan, numbers that leave the same remainder are treated as members of the same family."

Ant asks: "The gate wants remainder badge 3 modulo 11. Can more than one number have this badge?"

☐ Yes ☐ No

Gnam says: "Let me check by moving full cycles of 11."

$3 - 22 =$ _____, $3 - 11 =$ _____, $3 =$ _____, $3 + 11 =$ _____, $3 + 22 =$ _____.

Do all these numbers land at badge 3 modulo 11?

☐ Yes ☐ No

Look at the numbers above carefully.

What pattern do you observe?

Using this pattern, write the family of badge 3 modulo 11.

$\{\ldots, _____, _____, 3, _____, _____, \ldots\}$

Ant asks: "Instead of listing forever, can you write the whole family using one expression?"

Family of 3 modulo 11 = { _____ : _____ }.

Gnam says: "Now I think I can write this for any badge."
If the badge is a and the cycle length is n, first write the nearby family members:

$\{\ldots, _____, _____, a, _____, _____, \ldots\}$

Now write the full family in one expression.

Family of a modulo n = { _____ : _____ }.

Scene 4: Gnam's Difference Test

Gnam says: "Numbers in the same family look different, but maybe their difference tells us something."

Ant says: "Let us test a few pairs first."

The following pairs belong to the same family modulo 11. Complete the table.

Pair	Difference	Divisible by 11?
25, 3	$25 - 3 =$ _____	☐ Yes ☐ No
47, 14	$47 - 14 =$ _____	☐ Yes ☐ No
36, 25	$36 - 25 =$ _____	☐ Yes ☐ No
-8, 3	$-8 - 3 =$ _____	☐ Yes ☐ No

What do you notice about the differences?

Your Answer:

Ant asks: "Can we explain why this always happens?"
One member of the family of 3 modulo 11 can be written as

$$3 + 11p.$$

Write another member of the same family: _____
Now subtract the two members and simplify.

$$(3 + 11p) -$$

What number is always a factor in the difference?

Answer:

So, if two numbers belong to the same family modulo 11, their difference is divisible by _____.
Ant finally introduces the symbol for **"same cycle position."** Since 25 and 3 leave the same remainder when divided by 11, mathematicians write

$$25 \equiv 3 \pmod{11}.$$

This is read as: **"25 is congruent to 3 modulo 11."**

Ant says: "We have seen the idea through families and differences. Now let us write the same idea formally."

What does $a \equiv b \pmod{n}$ mean?

Let $a, b \in \mathbb{Z}$ and let n be a positive integer. We say that

$$a \equiv b \pmod{n}$$

if a and b leave the same remainder when divided by n. Equivalently,

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

Equivalently, $a = b + kn$ for some integer k.

Scene 5: The Modular Addition Game

Ant and Gnam enter Titan's playground.

For this game, only totals that appear after a player's turn are counted as previous totals.

1. Player A chooses a positive integer n.
2. Player B decides who goes first.
3. The running total T starts at 0.
4. On each turn, a player chooses any integer m and adds it to T.
5. A player loses if the new total is congruent to any previous total modulo n.

Understand the Game

Alice chooses $n = 5$. Bob goes first.

Round	1	2	3	4	5
Alice		-88		-23	
Bob	7		102		13
Running Total T	7	-81	21	-2	11
Remainder mod 5					

Bob loses because $11 \equiv 21 \pmod 5$. What number could Bob have picked instead of 13?

Your Answer:

Play the Game

Player A: _____ Player B: _____ Modulus n: _____

Round	1	2	3	4	5	6	7	8	9	10
Player A										
Player B										
Total T										
Remainder										

How many safe totals can appear before a remainder must repeat?

Your Answer:

Ant says: "In the game, only the final landing position mattered. Now let us see whether we can replace large numbers by smaller numbers before calculating."

Scene 6: Reduce First, Then Calculate

Gnam says: "These numbers may look big, but maybe we do not need to carry the full numbers."

Ant says: "Let us check whether smaller representatives give the same landing position."

Work modulo 11.
First reduce the numbers.

$$13 \equiv ____ \pmod{11} \qquad 27 \equiv ____ \pmod{11}$$

Now check addition in two ways.

$$13 + 27 = _____ \quad , \quad _____ \equiv ____ \pmod{11}$$

$$2 + 5 = _____ \quad , \quad 7 \equiv ____ \pmod{11}$$

So,

$$13 + 27 \equiv ____ + ____ \pmod{11}.$$

Let $a, b, c, d \in \mathbb{Z}$ and let n is a positive integer.
If

$$a \equiv b \pmod{n} \qquad \text{and} \qquad c \equiv d \pmod{n},$$

then

$$\boxed{a + c \equiv b + d \pmod{n}.}$$

Now, check subtraction in two ways.

$$27 - 13 = _____ \quad , \quad _____ \equiv ____ \pmod{11}$$

$$5 - 2_____ \quad , \quad _____ \equiv ____ \pmod{11}$$

So,

$$27 - 13 \equiv ____ - ____ \pmod{11}.$$

Let $a, b, c, d \in \mathbb{Z}$ and let n is a positive integer.
If

$$a \equiv b \pmod{n} \qquad \text{and} \qquad c \equiv d \pmod{n},$$

then

$$\boxed{a - c \equiv b - d \pmod{n}.}$$

Now, check multiplication in two ways.

$$13 \cdot 27 = _____ \quad , \quad _____ \equiv ____ \pmod{11}$$

$$2 \cdot 5 = _____ \quad , \quad _____ \equiv ____ \pmod{11}$$

So,

$$13 \cdot 27 \equiv ____ \cdot ____ \pmod{11}.$$

Let $a, b, c, d \in \mathbb{Z}$ and let n is a positive integer.
If

$$a \equiv b \pmod{n} \qquad \text{and} \qquad c \equiv d \pmod{n},$$

then

$$\boxed{ac \equiv bd \pmod{n}.}$$

Now multiply both sides by the same number.

$$13 \equiv ____ \pmod{11}$$

Multiply both sides by 4.

$$4 \cdot 13 = _____ \quad , \quad _____ \equiv ____ \pmod{11}$$

$$4 \cdot 2 = _____ \quad , \quad _____ \equiv ____ \pmod{11}$$

So,

$$4 \cdot 13 \equiv ____ \cdot ____ \pmod{11}.$$

Let $a, b \in \mathbb{Z}$ and let n is a positive integer.
If

$$a \equiv b \pmod{n},$$

then for every integer k,

$$\boxed{ka \equiv kb \pmod{n}.}$$

Now, raise both sides to the same power.

$$13 \equiv ____ \pmod{11}$$

Square both sides.

$$13^2 = _____ \quad , \quad _____ \equiv ____ \pmod{11}$$

$$2^2 = _____ \quad , \quad _____ \equiv ____ \pmod{11}$$

So,

$$13^2 \equiv ____ \pmod{11}.$$

Let $a, b \in \mathbb{Z}$ and let n is a positive integer.
If

$$a \equiv b \pmod{n},$$

then for every positive integer k,

$$\boxed{a^k \equiv b^k \pmod{n}.}$$

Now, check a chain of same-family numbers.

$$24 \equiv ____ \pmod{11} \qquad 13 \equiv ____ \pmod{11}$$

So,

$$24 \equiv 13 \pmod{11} \qquad \text{and} \qquad 13 \equiv 2 \pmod{11}.$$

Therefore,

$$24 \equiv ____ \pmod{11}.$$

Let $a, b \in \mathbb{Z}$ and let n is a positive integer.
If

$$a \equiv b \pmod{n} \quad \text{and} \quad b \equiv c \pmod{n},$$

then

$$\boxed{a \equiv c \pmod{n}.}$$

Now at another gate of Titan, the cycle length is 6. The guard shows Gnam a strange equality.

$$2 \cdot 1 = _____ \qquad 2 \cdot 4 = _____$$

Now check their landing positions on the modulo 6 cycle.

$$2 \cdot 1 \equiv ____ \pmod{6} \qquad 2 \cdot 4 \equiv ____ \pmod{6}$$

So we get

$$2 \cdot 1 \equiv 2 \cdot 4 \pmod{6}.$$

Gnam asks: "Since both sides have 2, can I cancel it?"
Can we write this?

$$1 \equiv 4 \pmod{6}$$

☐ Yes ☐ No

Check the landing positions of 1 and 4 on the modulo 6 cycle.

$$1 \equiv ____ \pmod{6} \qquad 4 \equiv ____ \pmod{6}$$

Are they the same?

☐ Yes ☐ No

So, can we always cancel a common factor in congruences?

☐ Yes ☐ No

Note

General Conclusion.
Let $a, b, k \in \mathbb{Z}$ and n is a positive integer, where n is the cycle length. Let $d = \gcd(k, n)$. Then

$$\boxed{ka \equiv kb \pmod{n} \iff a \equiv b \pmod{n/d}.}$$

In particular, if $\gcd(k, n) = 1$, then

$$ka \equiv kb \pmod{n} \implies a \equiv b \pmod{n}.$$

Thus, a common factor can be cancelled without changing the modulus when it is coprime to the modulus.

Practice Problems

Question 1. Find the remainder of 46^{100} when divided by 5.

Your Answer:

Question 2. A number is called **threeven** if it leaves remainder 0 when divided by 3, **throdd1** if it leaves remainder 1, and **throdd2** if it leaves remainder 2.

Complete the following modulo 3, and then name the type of the answer.

$$\text{threeven} + \text{throdd1} \equiv _____ \pmod 3 \quad \Rightarrow \quad \text{threeven} + \text{throdd1 is of type } _____$$

$$\text{threeven} + \text{throdd2} \equiv _____ \pmod 3 \quad \Rightarrow \quad \text{threeven} + \text{throdd2 is of type } _____$$

$$\text{throdd1} - \text{throdd2} \equiv _____ \pmod 3 \quad \Rightarrow \quad \text{throdd1} - \text{throdd2 is of type } _____$$

$$\text{throdd2} - \text{throdd1} \equiv _____ \pmod 3 \quad \Rightarrow \quad \text{throdd2} - \text{throdd1 is of type } _____$$

$$\text{threeven} \times \text{throdd1} \equiv _____ \pmod 3 \quad \Rightarrow \quad \text{threeven} \times \text{throdd1 is of type } _____$$

$$\text{throdd1} \times \text{throdd2} \equiv _____ \pmod 3 \quad \Rightarrow \quad \text{throdd1} \times \text{throdd2 is of type } _____$$

Explain one rule out of all.

Your Answer:

Question 3. Mary enters a sleep chamber at 6 pm. The chamber has a 12-hour clock. She sets the timer for 76 hours.

At what time will the alarm ring?

Your Answer:

Question 4. Sam starts eating green eggs and ham on Monday. Consider Monday as Day 1. He continues eating them for 200 days.

On which day of the week will he stop?

Your Answer:

Question 5. A 6×6 chessboard is coloured black and white in the usual alternating pattern. Two opposite corner squares are removed.

Can the remaining 34 squares be covered exactly by 17 dominoes, where each domino covers two adjacent squares?

Justify your answer.

Your Answer:

Question 6. Joanna starts by facing **west**. She rotates counterclockwise by $7110°$.

Cycle length for angles = _____

After reducing $7110°$ using this cycle length, what direction will Joanna face? Justify your answer.

Your Answer:

A Frightful Halloween



Question 7. On a frightful Halloween night, Todd is waiting outside a haunted house. His friends count the ghosts, but they give him only remainder clues.

Eileen counted the ghosts by 3's and found 2 left over.

Jane counted the ghosts by 5's and found 3 left over.

Eric counted the ghosts by 19's and found 7 left over.

Let x be the number of ghosts. Then

$$x \equiv 2 \pmod 3, \qquad x \equiv 3 \pmod 5, \qquad x \equiv 7 \pmod{19}.$$

What is the smallest possible number of ghosts?

Your Answer:

Question 8. Some soldiers are standing in a group.

Let x be the total number of soldiers.

When they line up in rows of 5, 2 soldiers are left over.

When they line up in rows of 7, 4 soldiers are left over.

The total number of soldiers is less than 100.

Find all possible values of x.

Your Answer:

Gnam: "So remainder is not just what is left after division. It is also a position in a cycle."

Ant: "Yes. Modular arithmetic is the mathematics of cycles."

Gnam: "And now big numbers do not look so big anymore."

Why Should an Engineer Care?

Modular arithmetic is one of the hidden tools behind modern computing and Artificial Intelligence.

- **Cryptography:** Secure communication on the internet (RSA, digital signatures, blockchain) relies heavily on modular arithmetic.
- **Computer Arithmetic:** Computers store numbers using a fixed number of bits. Overflow behaves like arithmetic modulo 2^n.
- **Error Detection and Coding:** QR codes, communication systems, and data storage use modular arithmetic to detect and correct errors.
- **Machine Learning Systems:** Large-scale AI software relies on efficient numerical computation, where modular ideas frequently appear in optimization, hashing, indexing, and distributed systems.

In short: whenever a system wraps around, repeats in cycles, encrypts information, hashes data, or performs efficient large-number computations, modular arithmetic is working behind the scenes.

End of Worksheet